Reverse Foundations of Physics

Separating Physical Postulates, Mathematical Carriers, and Logical Strength

OpenAI Codex* Asger Alstrup Palm†

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Abstract

What part of a physical theory is physics, what part is mathematics, and what part is inherited from the logic and existence principles in which the mathematics is developed? Familiar foundational programmes usually vary one of these coordinates while holding the others fixed. Reverse mathematics calibrates the axioms needed for a theorem. Reverse physics seeks physical postulates sufficient to recover a law. Operational reconstructions recover quantum structure from informational principles. Constructive, computable, choice-free, topos-theoretic, and finite approaches each replace selected pieces of the mathematical architecture. None of those methods alone answers the combined question.

We formulate a programme of *reverse foundations of physics* around the typed derivability judgement

$$L + S + M + \mathrm{Enc}(P) \;\vdash\; O.$$

Here L is the logic, S the set, type, or existence theory, M the mathematical carrier and analytic machinery, P the physical postulates after an explicit encoding, and O one sharply delimited obligation. The programme asks which ingredients are sufficient, necessary, equivalent, independent, or avoidable by reformulation. Its practical research atlas has three coarse axes—six foundational regimes, six carrier families, and sixteen physical obligations—giving 576 navigational coordinates. The cube is not an ontology and its coordinates are not presumed independent.

Five case studies show why the typing matters. An explicit mode labelling can avoid basis-selection uses of Choice without making physics "imply ZF." A Krein fundamental symmetry permits explicit normalized states but does not select a physical state. A represented scalar wave evolution is formalizable over RCA_0, while causal Green support remains a distinct theorem. Exact finite graph causality is not continuum Lorentzian causality. And a bounded finite BV contraction can be checked in PRA without lowering any infinite-dimensional quantum claim. These are sufficiency and separation results, not a universal weakest-foundation theorem. We close with a certificate-oriented research protocol and a set of tractable open problems at the interfaces between logic, analysis, and physics.

1 The missing question

A physical theory is normally presented as if its assumptions came in one list. In practice that list mixes several kinds of commitment. A symmetry principle, a causal condition, and a composition rule are physical postulates. A Hilbert space, a distribution space, a Krein product, or a C^*-algebra is a mathematical carrier. Completeness, compactness, selection of bases, spectral resolution, and

*Literature synthesis, formal organization, and drafting.

†Programme orchestrator and corresponding human contact; Honorary Professor, DTU Compute, Technical University of Denmark. asger@area9.dk.

existence of Green operators are mathematical theorems. Classical logic, set existence, actual infinity, and forms of Choice live one level further down. A calculation can use all four layers without saying where one ends and another begins.

This creates two opposite errors. The first is to treat the conventional mathematical formulation as physical content: quantum mechanics is said to *assume Hilbert space*, even when an operational reconstruction derives a Hilbert representation from other principles. The second is to treat the entire standard mathematical background as unavoidable: a proof uses a basis or compactness theorem, and one concludes that the physical law requires the Axiom of Choice. In both directions the inference skips the representation and the base theory.

Reverse mathematics supplies the right discipline for the mathematical side. Its central question is which set-existence axioms are needed to prove a theorem of ordinary mathematics [13]. Reverse physics applies a parallel strategy to physical laws: begin with accepted laws or theory structure and seek physical starting assumptions sufficient to recover them [5]. Operational reconstructions of quantum theory do something closely related inside a chosen formal arena [9, 6]. Yet the results do not compose automatically. A reconstruction may derive finite-dimensional complex Hilbert structure while relying on classical real analysis. A reverse- mathematical result may calibrate an abstract differential equation while saying nothing about which physical representation should encode its data. Computable analysis may construct an operator under a representation without classifying its proof-theoretic strength. A topos formulation may change the internal logic while leaving the bridge to experimental probabilities open.

The question of this paper is therefore not merely "which physical assumptions imply a theory?" nor merely "which axioms prove its mathematics?" It is:

> For a precisely represented physical obligation, which combination of physical postulates, carrier structure, existence principles, and logic is sufficient; which parts can be recovered from the obligation; and which can be removed by changing representation?

We use *reverse foundations of physics* as a working name for this combined programme. The phrase is descriptive, not a priority claim. The adjacent literatures are extensive, and the present literature survey is scoped rather than exhaustive. Our claim is narrower: the sources reviewed here do not provide a single typed atlas in which physical obligations, carrier choices, and logical or set-theoretic strength are all varied and audited together.

The programme grew from conformal gravity because that theory is a demanding stress test. Gauge symmetry, indefinite pairings, fourth-order evolution, Green operators, local BRST cohomology, state selection, renormalized products, and the quantum master equation all occur, but they occur at different dependency levels. The method is not specific to Weyl gravity. Its natural domain includes classical PDE, quantum mechanics, algebraic quantum theory, lattice systems, operational reconstructions, and any theory in which a change of mathematical foundations may change what "existence" or "state" means.

Contributions. This paper makes four contributions. First, it gives a typed formal object for the combined analysis. Second, it organizes a three-axis research atlas and explains what its coverage counts do and do not mean. Third, it presents five independently checkable case studies in which familiar foundational inferences split into distinct claims. Fourth, it proposes an evidence and certificate protocol designed to turn a suggestive matrix cell into a result that can survive changes of proof, representation, and implementation.

2 A typed derivability problem

2.1 Profiles, obligations, and encodings

The basic object is not a theory name but a derivability profile.

Definition 2.1 (Assumption profile)**.** An assumption profile is a tuple

$$\mathcal{A} = (L, S, M, \mathrm{Enc}(P), R),$$

where L specifies the logic and inference rules, S specifies the existence theory, M specifies the mathematical carrier and theorems made available over $L + S$, P is a list of physical postulates, Enc is a formal encoding of those postulates, and R records the representation of inputs and outputs.

The representation R deserves its own coordinate. A real number may be a Dedekind cut, a rapidly converging Cauchy name, or a point of an internal real object. A field may be a smooth section, a distribution, a Sobolev class, a finite lattice function, or a compatible family of finite data. Those representations can change both the content of an existence claim and the strength of its proof. We display R inside the profile but often absorb it into M and Enc, giving the compact judgement

$$L + S + M + \mathrm{Enc}(P) \ \vdash_R \ O. \tag{1}$$

Definition 2.2 (Obligation)**.** An obligation O is one theorem-level job with a declared result kind and scope: for example, construction of one normalized algebraic state; uniqueness of an energy solution in a represented carrier; existence of retarded and advanced Green maps with a support theorem; classification of local counterterms; or restoration of a local quantum master equation.

The obligation must be smaller than a theory. "Construct quantum field theory" is not an obligation. "Construct a BRST-compatible Hadamard state for the full metric BV complex" is an obligation, albeit a very difficult one. This granularity prevents a finite-mode calculation from being promoted to a continuum field theorem and prevents state existence from being promoted to a state-selection principle.

2.2 Seven relations, not one arrow

An unlabelled implication arrow hides too much. We use the following relation types.

Relation	Meaning
`USED_BY_DISPLAYED_PROOF`	One inspected proof invokes the item. This is neither necessity nor an optimal upper bound.
`SUFFICIENT_OVER_BASE`	A derivation of the obligation is available over the declared base, encoding, and representation.
`NECESSARY_OVER_BASE`	A reversal, separating model, or counterexample proves that the item cannot be omitted over the declared base.
`EQUIVALENT_OVER_BASE`	Both sufficiency and necessity have been proved over the same base and encoding.
`AVOIDED_BY_REFORMULATION`	Another explicit representation reaches the same declared obligation without the item.
`INDEPENDENT_OVER_BASE`	Models or realizations on both sides establish independence over the named base.
`UNKNOWN`	No stronger relation is certified.

These types separate three questions that are often collapsed:

1. Did a proof use a principle?
2. Is the principle sufficient or necessary for the theorem over a named base?
3. Does the physical obligation require the theorem in that representation and generality?

Only the second question is a reverse-mathematical calibration. The third is where reverse mathematics and reverse physics meet.

2.3 Why physical assumptions do not simply imply Choice

The phrase "physical assumption P implies the Axiom of Choice" is usually ill-typed. A physical postulate and a set-theoretic sentence do not even inhabit a shared formal language until an encoding and base theory have been fixed.

Proposition 2.3 (Encoding discipline)**.** *Let P be a physical postulate stated in an experimental or operational vocabulary and let C be a choice principle stated in a foundational language. A claim $P \Rightarrow C$ has no formal derivability content until one supplies a common theory $L + S$, an encoding $\mathrm{Enc}(P)$, and a target translation of C. Any proved implication then belongs to that encoding and base; it is not automatically an implication from an experiment to an unrestricted assertion about all sets.*

This does not make such reversals impossible. A physical principle may be encoded as a totality, selection, compactness, or continuity statement, and that statement may turn out to be equivalent to a familiar subsystem or choice fragment. The proposition says what must be provided before the claim is meaningful.

Physical structure can nevertheless reduce mathematical commitments in a more direct way. If the physics supplies a canonical grading and explicit mode labels, a proof need not choose a basis. If it supplies a preferred state or boundary condition, a selection problem may disappear. The correct relation in such cases is often `AVOIDED_BY_REFORMULATION`, not "physics proves Choice" or "physics disproves Choice."

3 The research atlas

The formal profile has many coordinates. For navigation we project it onto three axes: foundational regime, carrier, and physical obligation. The current projection has $6 \times 6 \times 16 = 576$ coordinates. This number is large enough to expose blind spots and small enough to inspect. It is not a claim that the underlying space of theories is finite, that the three axes are independent, or that every coordinate is coherent.

3.1 Six foundational regimes

Regime	Intended question
Classical standard	What is known in ordinary classical ZFC-style analysis, where background Choice is often not audited?
Weak arithmetic	How much proof-theoretic strength is needed in systems such as PRA, RCA_0, WKL_0, or ACA_0?
ZF with weakened Choice	Which constructions survive in classical set theory when full Choice or Countable Choice is absent or isolated?
Constructive/computable	What witnesses, algorithms, moduli, or represented operators can be constructed without treating classical existence as enough?
Topos/internal	What becomes of the theory when objects and truth are interpreted internally, often with Heyting rather than Boolean logic?
Finite/discrete restriction	What survives on finite carriers, exact finite algebras, lattices, or proposals that restrict actual infinity?

These regimes overlap but are not synonyms. ZF without Choice retains classical logic and actual infinity. Bishop-style constructive mathematics changes the meaning of existence. Computable analysis fixes representations and algorithms but is not by itself a subsystem classification. A topos has an internal logic relative to a selected category. A finite cutoff inside ZFC is not finitism.

3.2 Six carrier families

Carrier	What it is meant to track
Finite exact algebra	Finite matrices, integer or rational complexes, and explicit finite-dimensional algebraic data.
Hilbert/operator	Positive Hilbert spaces, completions, operator domains, spectral data, and one-parameter groups.
Krein/indefinite	Indefinite pairings, fundamental symmetries, positive companion topologies, and their operator theory.
Algebraic C^*-system	Observable algebras, states, GNS representations, nets, and algebra-first dynamics.
Smooth/PDE/ distributional	Manifolds, sections, Sobolev and distribution spaces, differential operators, and Green theory.
Localic/synthetic/ internal	Locales, internal algebra objects, formal manifolds, and synthetic smooth structures.

The rows are deliberately not mutually exclusive descriptions of a complete theory. A Krein space normally carries a positive companion Hilbert topology; an algebraic quantum theory may be represented on a Hilbert space; and a PDE may be encoded in a finite exact approximation. The axis asks which carrier is doing the work in the obligation under inspection.

3.3 Sixteen obligations in four groups

The obligation axis is the most important because it blocks silent promotion. For readability the sixteen values fall into four groups.

Group	Obligations
Kinematics and states	kinematics/observables; state existence; state representation; probability rule; physical state selection.
Dynamics and causality	generator/spectral dynamics; evolution and well-posedness; causal propagation and advanced/retarded Green maps.
Gauge and interaction	gauge/BV/cohomology; interaction construction; counterterm classification; anomaly classification.
Quantum completion and comparison	renormalized products; QME restoration; residual quantum transfer; reconstruction and continuum or empirical limits.

This ordering encodes no claim that every theory must pass through the stages linearly. It does show why "we have a state" does not answer which state is physical, why a unitary group does not supply a retarded propagator, and why a local anomaly classification does not restore a quantum master equation.

3.4 Coverage is metadata, not truth

In the current evidence snapshot, 452 of the 576 coordinates are emitted for inspection. Among those, 93 are marked as literature results, 88 as local results, 160 as having only pieces, 30 as priority gaps, and 81 as reviewed but not mapped. The remaining 124 are synthetic complements not yet emitted as coherent coordinates. Thus 205 positions lack substantive coverage in the current explorer. The evidence interface contains 69 source or certificate records.

Those numbers measure the state of a corpus and classification scheme. They do not measure how much humanity knows, and an unfilled cell is not evidence of impossibility. A literature-result label says that a source establishes a claim with a recorded boundary; it does not say the entire cell is solved. A local-result label says that a repository certificate reaches the declared obligation; it does not make the result representation-independent. A priority gap is an invitation to research, not a no-go theorem.

The counts are reproducible from `FOUNDATIONAL_INTERSECTION_CUBE_V4` and `FOUNDATIONAL_MATRIX_EXPLORER_SITE_V2`. The cube explicitly records that it is not literature complete, does not assess all 576 coordinates, proves no weakest base or reversal for the coded wave result, and establishes no new Lorentzian Weyl claim.

4 How adjacent programmes fit—and fail to collapse

4.1 Reverse mathematics

Reverse mathematics most directly calibrates the $L + S \vdash T$ part of Eq. (1). Its strength is the insistence on a named weak base and, in a full reversal, proofs in both directions. Simpson's analysis of the Cauchy–Peano theorem is an instructive physics-adjacent example: over RCA_0, the theorem is equivalent to WKL_0, while the Ascoli lemma and Osgood's maximum-solution theorem have the strength of ACA_0 [12]. "Differential equations require analysis" is thereby replaced with several exact statements whose strength depends on the theorem.

The limitation for our purpose is equally clear. Reverse mathematics begins after a statement and its representation have been selected. It does not by itself decide whether the physical problem requires Cauchy–Peano in that generality, whether a canonical physical flow avoids the compactness step, or whether a different carrier expresses the same observable content.

4.2 Reverse physics and operational reconstruction

Reverse physics begins from laws and seeks physical assumptions sufficient to rederive them [5]. This is the closest methodological relative of the present programme. Our additional demand is to audit the mathematical and logical substrate of every displayed derivation.

Quantum reconstructions show why this matters. In Hardy's five-axiom derivation, continuous reversible transformations are load-bearing: deleting continuity yields classical probability theory within the stated framework [9]. Chiribella, D'Ariano, and Perinotti define a broad operational-probabilistic class using five axioms and use purification to single out quantum theory [6]. These works turn some textbook mathematical postulates into derived representation theorems. They do not, without a separate audit, classify the proof-theoretic cost of the continuum, compactness, convex separation, or infinite-dimensional limit.

Accordingly, "Hilbert space is derived" and "Hilbert-space quantum field theory is constructible over a weak or choice-free base" are very different claims. The former can be a finite-dimensional operational theorem; the latter includes completion, unbounded operators, distribution theory, locality, states, and renormalization.

4.3 Choice-free functional analysis

It is common to summarize the situation as "functional analysis needs Choice." Recent results make that too coarse. Blackadar, Farah, and Karagila study Hilbert spaces and operators in ZF without Countable Choice, showing both robust canonical constructions and unfamiliar pathologies [4]. Blackadar and Farah develop substantial separable C^*-algebra theory in ZF, including representation theorems, spectral mapping for polynomials, and continuous functional calculus for commuting normal elements, while also exhibiting failures outside protected settings [3].

The lesson is not that Choice is irrelevant. It is that the cost is attached to a theorem, a class of objects, and a presentation. Canonically presented separable structures may avoid selections that arise for arbitrary families. The right unit of analysis is therefore not "Hilbert space" or "C^*-algebra" but the exact construction used in a physical obligation.

4.4 Constructive, computable, and proof-mined analysis

Constructive analysis requires existence statements to carry witness content [2]. Computable analysis makes that demand relative to explicit representations of spaces and operators [14]. Weihrauch and Zhong, for example, give an algorithm for distributional fundamental solutions of constant-coefficient differential operators under canonical representations [15]. That result is highly relevant to the carrier and computability axes, but a fundamental solution is not automatically a retarded or advanced solution and the theorem does not by itself supply Lorentzian causal support.

Proof mining asks a different question again: what quantitative or computational information can be extracted from an apparently nonconstructive proof? Pischke's metatheorem for nonlinear semigroups generated by accretive operators illustrates its reach into PDE and evolution, including extracted rates [11]. Proof mining should not be relabelled as a Big-Five reverse-mathematical classification, nor should a computable realizer be silently treated as a constructive proof in every foundational system. The three methodologies can inform one another only after their input and output types are recorded.

4.5 Topos and localic reformulations

Constructive Gelfand duality replaces point-set spectra with point-free spectra in a form valid constructively [7]. Heunen, Landsman, and Spitters place a noncommutative algebra in a topos of its commutative contexts, where the internal algebra is commutative, its spectrum is a locale, and the internal propositional logic is intuitionistic [10]. This is a genuine change of mathematical architecture, not merely deletion of Choice from a classical proof.

It also illustrates a recurring boundary. Internal state objects, locale valuations, and internal dynamics do not automatically supply a physical state-selection rule, a Lorentzian Green operator, an interacting BV complex, or empirical equivalence with the external formulation. Object logic and metalogic must be recorded separately. A non-Boolean algebra of quantum propositions is not, just by itself, an intuitionistic metatheory.

4.6 Finite mathematics and finite physics

Finite structures offer exact computation and sometimes eliminate analytic existence questions. But "finite" has several types. Gibbons, Hoffman, and Wootters use a finite field to label a discrete phase space when the quantum state-space dimension is a prime power [8]. The carrier of state vectors remains an N-dimensional complex Hilbert space. A finite field phase space is therefore not the same thing as a finite-field amplitude theory, a finite-mode truncation, or a rejection of actual infinity.

The distinctions become decisive when a finite regulator is used to motivate a continuum claim. Exact solvability at each finite size supplies a family of finite theorems. A continuum limit requires its own representation, uniform estimates, convergence theorem, and physical comparison. Finitism is a foundational proposal; a lattice cutoff inside ordinary mathematics is an approximation scheme.

5 Five certified separations

The following examples are not offered as a complete theory. They are small enough to verify independently and together show what the combined programme changes in practice.

5.1 Krein structure changes the pairing, not all Hilbert mathematics

A Krein space carries a nondegenerate indefinite product together with a fundamental decomposition or fundamental symmetry. If $J = J^* = J^{-1}$, the companion product

$$(x, y)_J = [x, Jy]$$

is positive and supplies Hilbert topology. Replacing a positive physical pairing by a Krein pairing therefore does not remove completion, topology, or operator-domain questions. It redistributes them.

In the reduced conformal-gravity mode carrier audited in `FOUNDATIONAL_KREIN_EXPLICIT_J_ZF_V1`, energy, family, chirality, and a finite multiplicity coordinate name every mode. The fundamental symmetry is coordinate multiplication by the sign of the family. At every fixed cutoff, the list and the involution are primitive-recursive finite data; PRA suffices to verify self-adjointness and $J^2 = 1$. At the complete one-particle level, the carrier is an explicitly indexed $\ell^2(I)$, and the occupation number Fock carrier is $\ell^2(\mathrm{Occ}(I))$. The certificate uses ZF results for those canonical completions and no Countable Choice.

The foundational work is done by the representation. Named modes replace an unspecified basis search and a choice of bases across arbitrary families. The certified relation is `AVOIDED_BY_REFORMULATION` for those particular selection steps and `SUFFICIENT_OVER_BASE` for ZF. It is not a proof that arbitrary Krein spaces have choice-free fundamental decompositions or that ZF is the weakest possible base.

This case also corrects the slogan "Krein throws out Hilbert space." The distinguished physical pairing becomes indefinite, but a positive companion Hilbert topology remains part of the displayed structure. What changes are the adjoint relations, positivity problem, null directions, and physical interpretation; what survives must still be audited theorem by theorem.

5.2 State existence is not state selection

Once the coordinate Krein carrier is available, constructing *a* state is easy. A named J-eigenvector e_i defines an explicit normalized vector state on the companion Hilbert algebra,

$$\omega_i(A) = (e_i, Ae_i)_J,$$

and the rank-one density operator $|e_i\rangle\langle e_i|$ is explicit. No choice principle is required for this witness. The same holds for named occupation vectors in the Fock carrier. These constructions are certified by `FOUNDATIONAL_KREIN_STATE_SELECTION_ZF_V1` with dependency tags `LOCAL-ALGEBRAIC` and `REDUCED-MODE`.

But J does not choose a unique coordinate. Indeed, the certificate proves a scoped no-selection result: no density operator is invariant under the full group of sign-preserving finite-coordinate permutations on the infinite positive and negative sectors. The reason is elementary. Invariance makes all diagonal weights within an infinite sector equal; trace-class normalization then forces each equal weight to vanish, contradicting unit trace. This excludes normal density states with that full invariance. It does not exclude singular states, and it does not prove that the permutation group is a required physical symmetry.

Thus three cells that are often merged must be separate:

$$\text{state existence} \quad \not\Rightarrow \quad \text{canonical state selection} \quad \not\Rightarrow \quad \text{Born interpretation.}$$

Extra physical information—dynamics, boundary conditions, thermality, Hadamard regularity, detector preparation, or an incoming condition—must do the selecting. The algebra and its set-theoretic foundation cannot supply that information merely by existing.

A related audit, `FOUNDATIONAL_BT_SEPARABLE_STATE_CHAIN_ZF_V1`, reaches the same boundary from an algebra-first direction. An explicitly presented separable compact- operator unitization, concrete states, and a corner GNS representation can be constructed in ZF. A semifinite trace is not a normalized state, and conditioning on a detector projection imports that projection as additional physical data. Algebra, state existence, representation, and physical selection are four obligations.

5.3 Weak wave evolution is not causal Green theory

The certificate `FOUNDATIONAL_CODED_POLYGONAL_WAVE_RCA0_V1` treats a mean-zero scalar wave on a coded circle. Dense states are pairs of rational periodic step functions whose polygonal primitives encode left- and right-moving profiles. Completed states and real times are supplied as fast Cauchy names with fixed rates. Rational translation is exact and preserves the energy metric. A finite-code time modulus and a primitive-recursive diagonal construction extend the translation group to named real times.

For this representation, RCA_0 suffices for existence, uniqueness, the group law, and energy conservation of the continuous isometric extension. Finite rational-code identities are already checkable in PRA. This is a genuine weak-base upper bound, and it demonstrates how much can be obtained when convergence rates are data rather than objects to be extracted from bare convergence.

The boundary is as important as the theorem. The result proves neither that RCA_0 is necessary nor that the upper bound is representation-invariant. It constructs no spacetime distribution, no advanced or retarded Green map, and no finite-propagation theorem. A unitary or isometric evolution group solves a Cauchy obligation; causal support is a separate spacetime theorem.

The literature makes the missing bridge visible. Normally hyperbolic operators on globally hyperbolic spacetimes have advanced and retarded Green operators with support control in classical analysis [1]. Under represented hypotheses, parts of PDE and fundamental-solution theory are computable [15]. The current audit has not found a direct reverse-mathematical reversal, a Bishop-style global Green theorem, or a choice-free ZF audit for the full variable-coefficient Lorentzian construction. "Classically known," "computably represented," and "calibrated over a weak base" are three different statuses.

5.4 Exact finite causality is not continuum causality

On a finite nearest-neighbour graph, a discrete wave recurrence can be solved by exact rational arithmetic. The retarded response at time step n is a polynomial in the finite propagation operator of degree at most n. Consequently its matrix entry vanishes when graph distance exceeds n. The advanced kernel follows by time reversal. The construction and graph-step support theorem are certified in `FOUNDATIONAL_FINITE_GRAPH_WAVE_CAUSALITY_V1`.

This is a clean causal theorem, but its type is `LOCAL-ALGEBRAIC/REDUCED-MODE`, not `LORENTZIAN-CAUSAL`. It does not provide a continuum Green operator, a light-cone support theorem, stability under refinement, a regulator- independent limit, or a metric BV propagator. Each of those requires new mathematical data.

The example gives a useful rule for finite programmes:

$$\begin{gathered}\text{exact finite identity} + \text{uniform estimate} \\ + \text{represented convergence} + \text{limit identification} \\ \implies \text{candidate continuum theorem.}\end{gathered}$$

Only the first term is supplied by exact finite causality. This does not make the finite theorem unimportant. It tells us exactly which part of continuum causality is algebraic and which part is analytic.

5.5 A finite BV contraction does not freeze the quantum theory

The energy-two free BV block provides another finite exact case. The certificate `FOUNDATIONAL_FREE_BV_ENERGY2_PRA_SDR_V1` verifies integer matrices for a strong deformation retract and checks the required identities in PRA. Because the contraction and projection are supplied explicitly, no Hahn–Banach extension or abstract complement selection is needed for this fixture.

The result is intentionally small. It does not verify all energies, the field-derived infinite complex, the Fock lift, a Green operator, a state, or a renormalization construction. It does not pass the classical import freeze gate for the quantum programme. The dependency tag is exactly `LOCAL-ALGEBRAIC`. A bounded exact certificate can remove foundational ambiguity from one algebraic step while leaving every analytic and causal obligation open.

This is the intended interaction with the Weyl BV–BFV programme. The classical complex is authoritative, but its import into quantum work must be checked object by object. Local counterterm and anomaly classification must precede coefficient calculations; restoration of the local quantum master equation must precede residual transfer. No reduced-mode or Euclidean result may be used as evidence for a Lorentzian causal claim. The reverse- foundational atlas makes these dependency cuts visible before a calculation is promoted.

6 What the case studies teach

The examples support several general lessons. They are methodological conclusions from the displayed certificates, not universal theorems about all physical theories.

6.1 The first cost of infinity is often completion, not Choice

In explicit separable carriers, countable infinity can enter through a named index set and canonical completion without any selection from an arbitrary family. The finite-to-infinite boundary still adds real mathematical content: summability, completion, operator domains, and limits. It simply need not add Countable Choice in the same step. "Infinite" and "choice-dependent" are not synonyms.

6.2 Physical input often acts as data, not as a foundational axiom

A mode grading, detector projection, boundary condition, or convergence modulus can remove an existential search. The physical postulate does not thereby become a set-theoretic axiom. It supplies a witness inside the chosen foundation. This is one of the most promising ways physics can lower the mathematical strength of a formulation: replace arbitrary existence with canonical physical data.

6.3 Representation sensitivity is scientific content

The RCA_0 wave theorem works because fast Cauchy rates and rational polygonal dense data are part of the representation. That is not a defect to hide. It tells us precisely which information makes the evolution effective and weakly formalizable. A later theorem may show that another natural representation is equivalent over the same base, or it may expose a genuine difference. Until then, representation invariance is an open obligation.

6.4 Positive structure migrates rather than disappears

Krein geometry removes positive definiteness from the distinguished pairing but restores a positive companion topology through a fundamental symmetry. Algebra-first formulations remove state vectors from the primitive vocabulary but recover Hilbert representations through GNS. Operational reconstructions remove Hilbert space from the primitive postulates but recover it from continuity, composition, and informational principles. In each case the assumption load moves. An honest analysis follows it.

6.5 The difficult cells are interfaces

Finite algebra, explicit coordinate carriers, and named states are relatively tractable. The sparse region begins where several interfaces must be crossed at once: weak or constructive analysis plus variable-coefficient Lorentzian PDE; internal topos structure plus external experimental probability;

gauge cohomology plus microlocal renormalization; or an operational reconstruction plus infinite-dimensional continuum limits. These gaps are not random. They mark where entire mathematical toolchains must be retyped.

7 A research protocol

The matrix becomes scientifically useful only if filling a cell means more than attaching a citation. We propose the following protocol.

7.1 Declare the target before auditing the proof

Every project begins with one obligation, one carrier, one representation, and one dependency boundary. "Wave existence" should say whether the solution is a Cauchy name, a Sobolev function, a distribution, or a smooth field. "State" should say algebraic state, normal density state, KMS state, Hadamard state, or physical selector. "Causality" should say graph-step support, finite propagation, or advanced/retarded Green support.

7.2 Build a proof ledger

The displayed proof is factored into individual dependencies: finite algebra, completion, compactness, extension, subsequence selection, spectral calculus, operator-domain closure, Green existence, support, duality, and so on. Each dependency gets one of the seven relation types. A proof use is never promoted to necessity.

7.3 Search for avoidance before proving a lower bound

Before attempting a reversal, ask whether the physical representation supplies a witness or canonical enumeration. An explicit diagonal operator may avoid a general spectral theorem. A named mode basis may avoid a basis-selection principle. Chiral coordinates may turn a PDE evolution into translation. This often produces a useful upper bound quickly and exposes the precise residue for a later lower bound.

7.4 Separate producer and checker

Exact finite claims should carry machine-readable witnesses and an independent checker that does not merely rerun the producer. Analytic claims should pin the theorem statement, hypotheses, representation, and source. Every result records what would falsify it, what was not checked, and which stronger claim it does not establish. Exact and numerical evidence remain distinct types.

7.5 Promote only through typed gates

For quantum Weyl work, the dependency tags

`LOCAL-ALGEBRAIC`, `EUCLIDEAN-SPECTRAL`, `REDUCED-MODE`, `LORENTZIAN-CAUSAL`

are noninterchangeable. Likewise the lifecycle states

`CLASSIFIED`, `COEFFICIENT_COMPUTED`, `QME_RESTORED`, `RESIDUAL_TRANSFERRED`,
`LORENTZIAN_CERTIFIED`

must not be collapsed. The atlas should show missing typed edges, not draw an unlabelled arrow around them.

8 Near-term research programme

The most valuable next problems are not simply the emptiest cells. They are cells for which one can isolate a small theorem that connects two mature literatures.

8.1 Weak localized wave identities

The represented Cauchy evolution over RCA_0 is established, while spacetime localization is not. A tractable next target is a weak integrated wave identity for compactly supported rational polygonal tests, followed by a support statement for a carefully coded (1+1)-dimensional model. This would test exactly where the move from energy evolution to spacetime distributions adds strength, without beginning with the full metric BV propagator.

8.2 A ZF audit of one Green construction

Choose one normally hyperbolic scalar operator on an explicitly presented globally hyperbolic spacetime. Factor the classical proof into Sobolev completion, energy estimate, local solution, exhaustion, uniqueness, duality, and support. For each step determine whether the standard proof merely uses Choice, whether a separable canonical formulation avoids it, or whether a choice fragment is genuinely needed. The first deliverable should be a dependency ledger, not a claim about all hyperbolic PDE.

8.3 Proof strength of an operational reconstruction

Select one finite-dimensional reconstruction, preferably the continuity branch in Hardy or the purification branch of Chiribella–D'Ariano–Perinotti. Formalize the exact real-algebraic and compactness principles used. Then separate the finite theorem from any proposed countable or field-theoretic extension. This is a promising place to obtain an actual equivalence or avoidance theorem because the physical postulates are already explicit.

8.4 Internal BV and the external probability bridge

The topos literature supplies internal commutative algebra, locale spectra, state representations, and versions of dynamics. A useful Weyl target is smaller than "construct QFT in a topos": internalize a finite local BRST complex and verify nilpotency and a finite contraction, then state precisely what additional structure would be required for integration modulo boundary terms, anomaly classification, and external probabilities. Even an obstruction ledger would convert a vague gap into typed missing morphisms.

8.5 Physical selectors on explicit state spaces

The state-existence problem is already easy in several explicit ZF carriers. The next scientific question is which physical assumptions select a state. Candidate selectors include a Hamiltonian ground condition, KMS condition, incoming detector preparation, residual symmetry, or a local Hadamard condition. Each should be tested independently for existence, uniqueness, foundational strength, and compatibility with gauge structure. A no-selection theorem is valuable only when its symmetry and state class are physically justified.

9 Limits of the present paper

This paper introduces a programme and reports bounded examples. It does not establish a global metatheorem relating physical postulates to set-theoretic strength. It does not prove that the three atlas axes form a literal Cartesian product of coherent theories. It does not prove that the six regimes, six carriers, or sixteen obligations are canonical or exhaustive.

The literature survey is a structured seed corpus, not a systematic review of all mathematical physics. "No reviewed source" means no source in the declared corpus, not that no result exists. The 576-cell count and coverage statistics describe a versioned research instrument. They are not novelty counts, evidence weights, or probabilities that a theorem is true.

The local foundational results are upper bounds unless a reversal is stated. The RCA_0 coded-wave theorem does not show RCA_0 is weakest. ZF sufficiency for the explicit Krein and separable C^* carriers is not a necessity theorem. Exact finite certificates do not settle infinite completions. Computable or constructive results are not automatically reverse-mathematical classifications.

Finally, none of the case studies constructs a complete Lorentzian off-shell BV propagator, a BRST-compatible Hadamard state for the full metric complex, renormalized Lorentzian time-ordered products, a causal perturbative AQFT, or a Lorentzian quantum-master-equation theorem. The certificates carrying `REDUCED-MODE` or `LOCAL-ALGEBRAIC` tags provide no evidence for those `LORENTZIAN-CAUSAL` obligations.

10 Reproducibility

The paper has a machine-readable claim map generated from current certificate inputs. It pins the source paths and hashes, extracts the atlas dimensions and status counts, checks the exact dependency tags of every local case study, and records the paper's nonclaims. The reproduction commands are:

```
python3 paper/generate_21_reverse_foundations_claim_map.py --check
python3 paper/verify_21_reverse_foundations_claim_map.py
pdflatex -interaction=nonstopmode -halt-on-error \
  paper/21-reverse-foundations-of-physics.tex
```

The principal machine-readable authorities are `FOUNDATIONAL_INTERSECTION_CUBE_V4`, `FOUNDATIONAL_MATRIX_EXPLORER_SITE_V2`, `FOUNDATIONAL_KREIN_EXPLICIT_J_ZF_V1`, `FOUNDATIONAL_KREIN_STATE_SELECTION_ZF_V1`, `FOUNDATIONAL_BT_SEPARABLE_STATE_CHAIN_ZF_V1`, `FOUNDATIONAL_CODED_POLYGONAL_WAVE_RCA0_V1`, `FOUNDATIONAL_FINITE_GRAPH_WAVE_CAUSALITY_V1`, and `FOUNDATIONAL_FREE_BV_ENERGY2_PRA_SDR_V1`.

11 Conclusion

The mathematical assumptions of a physical theory are neither inert notation nor a single hidden axiom. They are a structured dependency layer. Changing the carrier can move positivity, completion, and representation obligations. Changing the foundation can alter what counts as existence. Physical postulates can supply canonical data that avoid otherwise arbitrary choices. And the same physical slogan can have different strength under different encodings.

The practical response is to replace undifferentiated questions—"does physics need Choice?", "can Krein replace Hilbert space?", "can finite mathematics recover the continuum?"—with typed judgements of the form

$$L + S + M + \mathrm{Enc}(P) \ \vdash_R \ O,$$

where every symbol is explicit and every arrow has a relation type. The first case studies already change the picture. Explicit infinity need not introduce Choice at its first step. State existence does not select a physical state. Evolution does not establish causal support. Exact finite causality does not establish a continuum light cone. And a finite algebraic BV certificate does not promote an analytic quantum theory.

The resulting atlas is useful precisely because it contains holes. A blank cell is not an absence theorem; it is a request for a well-typed obligation, an encoding, a proof ledger, and an independently checkable result. Filling those cells one theorem at a time offers a realistic route toward understanding not only which physics follows from which physical assumptions, but which mathematics the physics truly needs.

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